Capital structure arbitrage: method guide

Merged from your files, Yu (2005) and Finkelstein (2002)

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The answer

You have three related methods, at three stages.

1. **Method A, your workbooks: rebuilt and tested.** It runs on CDS and reproduces your master workbook. Its gross Sharpe ratio of 2.51 has no hold-out, and it falls to 1.47 with a one-week lag and to 0.62 at 5% of spread with real turnover.
2. **Method B, Yu adapted to bonds: coded, not tried on real bonds.** The formulas reproduce CreditGrades Table C.1. The backtest loop has run only on made-up data.
3. **Method C, Finkelstein across bonds, options and stock: planned, not built.** The deck gives a model and hedges, but no trading rule. Every signal and trade in the plan is an addition.

Recommended next step: build work package WP1 of Method C. It needs no market data, and it also underpins the options leg that Methods A and B lack.

The main risk is the same in all three: nothing shows that this makes money after costs on bonds. Treat every number below as a test target, not as an expected return.

1. How the methods differ

	A. Your workbooks	B. Yu on bonds	C. Finkelstein
Instruments	CDS and stock	Bond and stock	Bond, options and stock
Fair value from	A weekly fit across names: ln(Imp/Hist) on ln(S/D)	Each name against its own model spread	One model fitted to options, then priced on the bond
Volatility	Blend of realised and put vols	1,000-day realised	Option term structure
Barrier $\bar{L}$	Fixed 0.5	Fitted on first days, then fixed	0.5 by default; skew only if it clearly fits better
Debt per share	CreditGrades Appendix B	Total liabilities over shares	Total liabilities over shares by default; the deck gives no formula
Recovery	0.40	0.50	0.40 proposed; deck uses 0.30 and 0.10
Rate	5-year swap	5-year Treasury	Treasury curve
Signal	Log gap to fair vol × 30, beyond ±6	Market beyond $(1+\alpha)\times$ model, $\alpha = 0.5$	Not in the deck. Proposed: monthly rank on the bond-stock gap, then two gaps
Hedge	Delta, asset vol held; weekly	Delta, held fixed at entry	Delta, plus puts for jump risk
Exit	None: book rebuilt weekly	Convergence, 180 days, or value below zero	Not in the deck. Proposed: by time, not convergence
Status	26 tests pass, 1 slow test skipped	161 tests pass	Check script only

2. Method A: your workbooks

Full detail, with cell references and example rows: `own_models/05_io_spec.md`. Code: `own_models/e2c_model.py`. Run everything with `run_pipeline(inputs, params)`.

Inputs: one row per name per week, 26 columns.

- Identity: date, ticker, sector.

- Balance sheet: short debt, long debt, other short and long liabilities, minority interest, preferred.
- Equity: price, market cap, realised vols at 30, 60, 90, 180 and 360 days, 162-week vol.
- Options, optional: put vols at 3, 6, 12 and 24 months.
- Credit and rates: 5-year CDS, 5-year swap.
- For P&L: next date, next spread, next price.

Step	Function	Takes	Gives
S01	`debt_per_share`	Balance sheet, market cap, price	Debt per share
S02	`average_equity_vol`	Realised and put vols	Blended equity vol
S03	`historical_asset_vol`	Equity vol, price, debt per share	Historical asset vol
S04–05	`survival_probability`, `model_spread`, `risky_annuity`	T, S, D, asset vol, L, lambda, R, r	Survival, spread, annuity, CDS value
S06	`implied_asset_vol`	CDS spread and the same inputs	Implied asset vol
S07	`leverage_regression`	ln(Imp/Hist), ln(S/D), each week	Weekly fit
S08–09	`fair_asset_vol`, `vol_residual`	Asset vol, leverage, fit, implied vol	Fair vol, residual
S10	`sector_adjusted_residual`	Residual, sector (optional)	Sector-adjusted residual
S11	`trade_signal`	Residual, spread	Long and short flags
S12	`cds_greeks`	Model inputs, spread	Annuity, delta, vega, rho, theta
S13–14	`trade_size`, `equity_hedge`	Flags, vega, delta, price	Sizes, hedge dollars and shares
S15	`weekly_pnl`	Sizes, hedge, next spread and price	Eight P&L lines
S16–18	`stress_tests`, `name_risk`, `closed_trade_flag`	Positions, greeks	Stress, risk, closed flag
S19–20	`book_capital`, `weekly_return`	Weekly book	VaR, margin, capital, NAV, return
S21	`bid_offer_cost`	Closed share, stress	Cost estimate, not deducted
S22	`performance_summary`	Weekly returns	Sharpe, mean, vol, NAV, t-stat, drawdown

Fixed values: T 5, L 0.5, lambda 0.3, R 0.4, residual scale 30, trigger ±6, target $200k on 3 vol points, trade cap $10M.

Not in your files, so you must choose: CDS and swap sources, vol estimator, put moneyness, universe rule, balance-sheet lag, execution lag, exit rule, dividends and borrow. The balance sheets are not point in time, and GM and Ford were removed because the model has no solution for them.

Target numbers in `12AdjE2C_Data_27_9_2005 (47.9MB, former employer folder).xls`, sheet `Reg`: `CK6` Sharpe 2.5106, `CM6` ending NAV 1.7992, `BA7` total P&L 20,477.8. I read these directly from the original file.

3. Method B: Yu on bonds and equity vol

Full detail, 97 steps: `paper_method/02_bond_vol_method.md`. Code: `paper_method/e2c_bond.py`. Run everything with `run_bond_backtest(panel, config)`. Yu's extraction is in `paper_method/01_paper_extraction.md`.

Inputs: issuer links and rating; bond terms; bond trade reports; share price, returns and dividends; total liabilities, shares and filing dates; Treasury curve; repo rates, cash rate and borrow fees. Optional: CDS, implied vol, event dates.

Part	Main functions	Takes	Gives
1	`bond_terms_are_eligible`, `pick_bond_and_sibling`	Issuer and bond terms, trade counts	One bond and a sibling
2	`debt_per_share`, `historical_equity_vol`, `zero_curve_from_par_yields`	Accounts, returns, par yields	D, equity vol, zero curve
3	`clean_trade_reports`, `daily_clean_price`, `z_spread`	Trade reports, bond, curve	Market Z-spread
4	`survival`, `risky_bond_price`, `model_z_spread`	Firm, bond, curve, recovery	Model price, model Z-spread
5	`fit_barrier_on_bond`	First days of market Z-spreads	$\bar{L}$, fit error, status
6	`measured_round_trip`, `planning_round_trip`	Trades or rating	Round-trip cost

Part	Main functions	Takes	Gives
7	`entry_signal`, `peer_adjusted_signal`, `sibling_confirms`, `implied_vol_bond`, `decide_entry`	Market and model Z, peers, sibling, vol	Signal, credit-implied vol, entry decision
8	`bond_equity_delta`, `rate_hedge_face`, `build_position`	Firm, bond, curve	Face, shares, Treasury hedge
9	`daily_pnl`	Position, marks, carry	P&L by leg
10	`bond_exit_reason`	Spreads, days held, value, events	Exit reason
11	`monthly_index`, `annual_sharpe`, `trade_summary`	Trades	Monthly returns, Sharpe, tables
12	`check_table_c1`, `check_no_lookahead`, `break_even_cost`	Panel, config	Pass or fail, break-even cost

Outputs: market and model Z-spread; $\bar{L}$ and fit status; credit-implied vol; each trade's side, dates, exit, sizes, return and P&L by leg; monthly returns, Sharpe, summaries, factor fit; break-even cost.

Warning from the paper. Yu's Equations 14–15 print $G(T)$ where $G(\xi)$ is needed. Copied as printed, they give an annuity of 5.31, above the riskless cap of 4.42. The code uses the corrected form.

4. Method C: Finkelstein on bonds, options and stock

Full plan, 23 pages: `finkelstein_plan/finkelstein_replication_plan.pdf`. Slide-by-slide extraction: `finkelstein_plan/01_deck_extraction.md`. No model code is built yet.

What the deck computes. The stock is a shifted lognormal with a knock-out, and default is asset value touching a barrier. Options set the asset vol term structure. The output is a closed-form survival curve, which gives bond and CDS values and splits each put into a default part and a diffusion part.

Inputs: stock price, dividends, shares, borrow fee; option quotes by strike and expiry, with open interest; bond terms, trades, rating, repo; balance sheet and filing times; Treasury yields and cash rate; event dates.

Outputs: survival curve; model bond price and Z-spread; default and diffusion parts of each put; option-implied default probability; greeks and jump to default; three market gaps and a trade decision.

Base signal, changed after reading Chen, Chen and Li (2026). The agent read that paper in full. It ranks names monthly on actual minus equity-implied bond spread, using CreditGrades with a payout drift, 3-year realised vol and no options. High-gap bonds outperform and high-gap stocks underperform. The bond premium is 5.28% a year (t = 5.64) over the full sample, and 3.24% (t = 2.69) on clean post-2002 data. The plan found no net-of-cost result in the paper. So trade T0, a monthly rank book, becomes the first trade to build, ahead of any fixed trigger. I have not checked these figures myself, because no copy of the paper is in your files.

The added idea: three markets give two independent gaps, $G_{bs} = G_{bp} + G_{ps}$. If the bond is out of line against both stock and options, trade the bond. If bond and options agree against the stock, the fixed vol is probably stale, so do not trade. This decision table is an addition, not Finkelstein's.

Pack	Produces	Needs	Done when
WP1	`fk_model.py` and tests	Nothing	Deck's model numbers pass, no market data
WP2	`fk_value.py`: bond, CDS, put split	WP1	Matches `risky_bond_price` at flat vol
WP3	`fk_risk.py`: greeks and hedges	WP1, WP2	Hedge tests pass
WP6a	`fk_des.py`: rank-book replication, first real run, no options	WP2	Signs match the paper; gross and net returns reported
WP4	`fk_options.py`: quotes to clean vols	WP1	Synthetic quotes round-trip
WP5	`fk_calibrate.py`	WP1, WP4	Fit test passes
WP6	`fk_gaps.py`	WP5, WP6a	Gaps run on a synthetic panel
WP7	`fk_trades.py`, `fk_backtest.py`	WP3, WP6	P&L legs sum to the account daily
WP8	`fk_american.py`	WP1	American price error stated
WP9	First option run on 20–50 high-yield names	WP4–WP7	Gaps, hedge slopes, and stability settled
WP10	Full backtest and report	WP8, WP9	Net results at 1, 2 and 3 times cost

Stop rules from the plan: stop if WP6a's net bond return at measured costs is zero or negative among large bonds, or if WP9's measured hedge slope is far from the model delta. A fixed trigger is a poor stop test: on GX the log gap is 0.02 to 0.08 on the deck's values, and 0.19 to 0.26 on the published settings.

Deck numbers, assuming a 5% rate. These reproduce: the GX skew table within 0.14 vol points, the barrier (6.01 fitted, 6 in the deck), the GX curve within 17 bp of chart readings, and all 12 hedge numbers within 7.2%.

These do not: the put prices at the printed 3% rate (I get 5.24 and 2.93, against 4.92 and 2.72, which fit a rate near 4.7–4.9%); the 1.42 default part (1.35 by the formula that gives 1.90); and slide 19's 22.2% vol, slide 10's asset vol stability, and slide 21's 422 bp at $\lambda = 0$.

5. What the tests across methods found

Tables and caveats: `scenarios/scenario_tables.pdf`. Real tests use your 2002–05 CDS panel, before costs and in sample. No real bond has been tested.

1. **Hedge size is the sharpest open conflict.** Your workbooks, CreditGrades and the Finkelstein deck all hold asset vol fixed, and the plan reproduces slide 21 that way. But on your real CDS, the smaller equity-vol delta cut variance by 7%, 16% and 17% at 1, 4 and 13 weeks, against −14%, 12% and 10% for the asset-vol delta. So: replicate with asset vol, and test bonds with both. Method C's WP9 measures this directly.
2. **Low-spread names are unreliable.** Under 100 bp, both deltas mostly add variance, and a +40 bp premium moves the delta by a median +59%. The first bond test leaves these names out. They were 79% of the 1-week CDS sample, so this removes most of an investment-grade universe.
3. **The liquidity checks have a blind spot.** Peer and sibling checks cut false trades in a made-up grid, but cannot see illiquidity across a whole issuer. On real CDS the peer check's benefit is not significant.
4. **A delta hedge barely covers default.** In the scenario table, it recovers about $1.4M of a $5.1M loss on $10M of bond. Finkelstein's put legs exist for this reason.
5. **Costs may decide the result.** A rough estimate puts bond round trips at 1–3% of capital, about the same as Yu's typical 1–3% trade return.

6. What is verified, and what is not

- **Rerun by me today:** Method A 26 pass and 1 skipped. Method B 161 pass. Scenarios 6 pass. The Finkelstein check script gives identical output. The agent reports the skipped slow test passing when enabled, and I separately read its target cells from the workbook.
- **Code matching a source is not evidence of profit.** Method A's tests prove that it matches the workbook. Method B's tests prove that it matches CreditGrades and is internally consistent.
- **Edited tests.** While revising, the Method B agent changed some code behaviour and edited 4 of its original 81 tests. No earlier copy exists, so I could not compare them. The exit, peer and barrier tests I read assert specific values and do not look loosened, but I cannot prove that.
- **Unread or lost:** the `.MDI` file, the missing put-implied-spread add-in, the lost Minimum MTM formula, the `AK SCRAP` files and the large archives, and the scanned SDS notes (read only in part).
- **Ownership.** Several files look like work from former employers. This use is personal research, but check your rights before taking any of it elsewhere.

7. Where everything is

Each method is on its own git branch, off `main`. Each branch has a checkout under `E2C/worktrees/`.

Branch	Folder	What it holds
`method/workbook`	`worktrees/workbook`	Method A: spec, code, tests, test vectors, panels
`method/yu-bonds`	`worktrees/yu-bonds`	Method B: paper extraction, 97 steps, code, tests, worked example
`method/finkelstein`	`worktrees/finkelstein`	Method C: plan PDF, deck extraction, check script
`compare/scenarios`	`worktrees/scenarios`	Scenario tables. Merges A and B, because it imports both
`docs/lit-and-guide`	`worktrees/docs`	This guide and the literature review
Not tracked	`evidence/`	The 55 downloaded files, the email index and the text of 21 emails